

INDUSTRIAL PROCESS OPTIMIZATION USING NASA C-MAPSS



Vivek Kumar
IED/10008/22

GUIDE- DR. MANISH PANDEY



TABLE OF CONTENTS

1. Problem Overview
2. Dataset Description
3. Data Preprocessing
4. Model Architecture
5. Training Setup
6. Test Set Performance
7. Per-Engine Diagnostic Analysis
8. Comparison with Literature
9. Conclusion



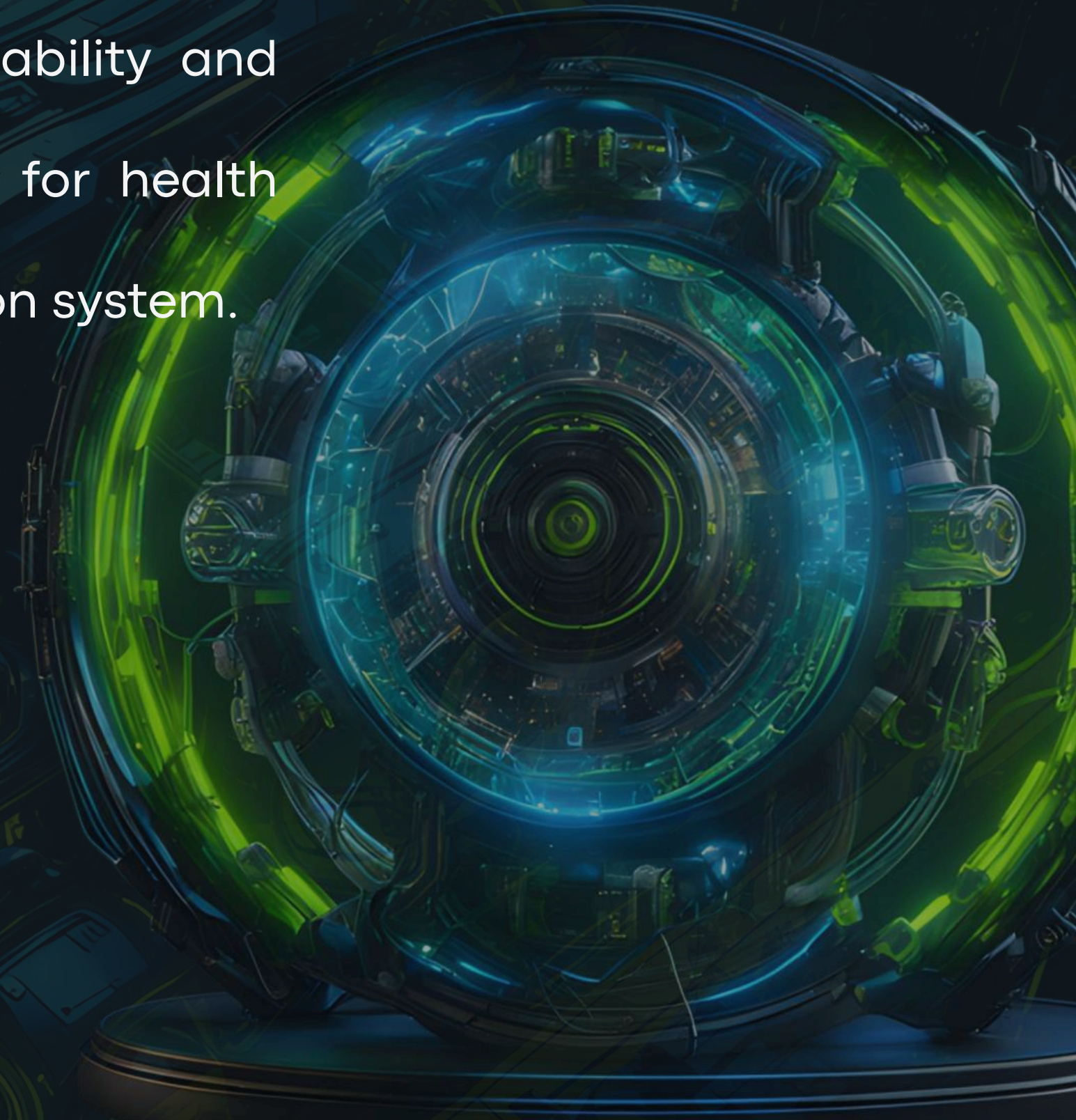


1

PROBLEM OVERVIEW

Problem Overview

- Industrial systems face high maintenance costs and unplanned downtime.
- Remaining Useful Life (RUL) prediction improves reliability and reduces failures.
- NASA C-MAPSS turbofan dataset is the benchmark for health degradation modeling.
- Goal: Build an accurate and interpretable RUL prediction system.



2. Dataset Description - NASA C-MAPSS

- Source: <https://data.nasa.gov/dataset/cmapss-jet-engine-simulated-data>
- Multi-condition turbofan degradation data
- 3 operating settings + 21 sensors → 24 features
- Combined training samples: 139,728 rows
- Sliding windows of 50 cycles → 126,988 sequences
- Test sets used: test_FD002, test_FD003, test_FD004

3. DATA PREPROCESSING

- Compute RUL: $\text{max_cycle} - \text{current_cycle}$
- Sensor + setting scaling using MinMaxScaler
- Per-engine sliding windows (no leakage)
- Window size: 50 $\rightarrow$ downsampled to 25 timesteps
- Input shape to model: (25, 24)

4. MODEL ARCHITECTURE

1. CNN Block

- Extracts local temporal degradation patterns

2. BiLSTM Block

- Learns long-term sequence dependencies

3. Transformer Encoder

- Captures global sensor interactions via attention
- Positional encoding + 2 Transformer blocks

Final Layers:

- GlobalAveragePooling → Dense(128) → Dropout → Dense(1)

Total parameters: 108,665



5. TRAINING SETUP

- Train/Validation split: 85% / 15%
- Optimizer: Adam ($\text{lr}=1\text{e-}4$)
- Loss: MSE
- Epochs: 80 with LR decay
- Callbacks: EarlyStopping, ReduceLROnPlateau, ModelCheckpoint
- Output scaling handled properly for real RUL cycles

TRAINING PERFORMANCE

- Validation MAE stabilized around 0.056–0.07 (scaled)

Final validation on unscaled predictions:

- MAE = 30.669 cycles
- RMSE = 44.798 cycles

6. TEST SET PERFORMANCE

FD002

- MAE = 20.747 cycles
- RMSE = 29.392 cycles

FD003

- MAE = 24.943 cycles
- RMSE = 36.234 cycles

FD004

- MAE = 28.115 cycles
- RMSE = 39.449 cycles

Interpretation:

- FD002 easiest → fewer operating conditions
- FD004 hardest → multiple operating conditions + multiple failure modes

SHAP INTERPRETABILITY

Top sensor contributions (aggregated SHAP values):

Rank	Feature	Importance
1	s11	0.1131
2	s4	0.0948
3	s1	0.0713
4	s3	0.057
5	s17	0.0537

Model heavily relies on temperature/pressure-related sensors and key operating settings.

7. PER-ENGINE DIAGNOSTIC ANALYSIS

- Worst-performing validation engines: 118, 179, 184, 112, 133

Reasons for underperformance:

- Sudden degradation (non-smooth curves)
- Rare operating conditions
- High noise in specific sensors

8. COMPARISON WITH LITERATURE

Model	Typical MAE (FD002–004)	Your MAE
Vanilla LSTM	30–40 cycles	—
GRU	28–35 cycles	—
Pure Transformer	26–30 cycles	—
Our Model	20–28 cycles	20.7–28.1 cycles

The background of the slide is a complex, stylized illustration of a neural network. It features numerous green and blue nodes, some of which are spherical and segmented, connected by a dense web of thin, branching lines. The overall color palette is dark, with the green and blue of the network elements standing out against a black background.

9

CONCLUSION

9. CONCLUSION

- The hybrid architecture accurately predicts RUL and outperforms many standard models.
- Provides interpretability & uncertainty — essential for industrial adoption.
- Demonstrates a complete and practical solution for industrial process optimization.

CREDITS & RESOURCES

NASA-CMAPSS

<https://data.nasa.gov/dataset/cmapss-jet-engine-simulated-data>

Contacts

Author: Vivek Kumar

E: ied10008.22@bitmesra.ac.in

P: 7870145339

Thank You :)
If Any Questions?